

High-Level Data Warehouse Design from an OLTP Movie Rental System

Course: Data Warehousing
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1. Introduction :

The movie rental business currently uses an OLTP (Online Transaction Processing) database system to support daily operational activities such as customer registration, movie rentals, inventory management, staff operations, and payment processing. While the OLTP database is efficient for handling transactional operations, it is not optimized for analytical reporting and business intelligence purposes.

Business managers require analytical capabilities to evaluate customer behavior, analyze rental trends, measure store performance, monitor revenues, and identify popular films and categories. Performing such analytical queries directly on the OLTP system can negatively affect operational performance due to the highly normalized structure of the database.

To address these limitations, a dimensional data warehouse design is proposed. The proposed design transforms operational data into an analytical environment optimized for reporting, trend analysis, and decision-making. This report presents a high-level dimensional model including fact tables, dimension tables, grain definition, star schema design, and ETL planning for the movie rental business.

2. Understanding the OLTP Schema :

The provided OLTP database schema represents a movie rental management system based on the Sakila database model. The schema is designed to support daily transactional operations including customer management, movie rentals, payments, inventory tracking, and staff activities.

The database contains multiple normalized tables connected through primary and foreign key relationships. This normalization improves transactional consistency and minimizes redundancy; however, it increases query complexity when performing analytical reporting.

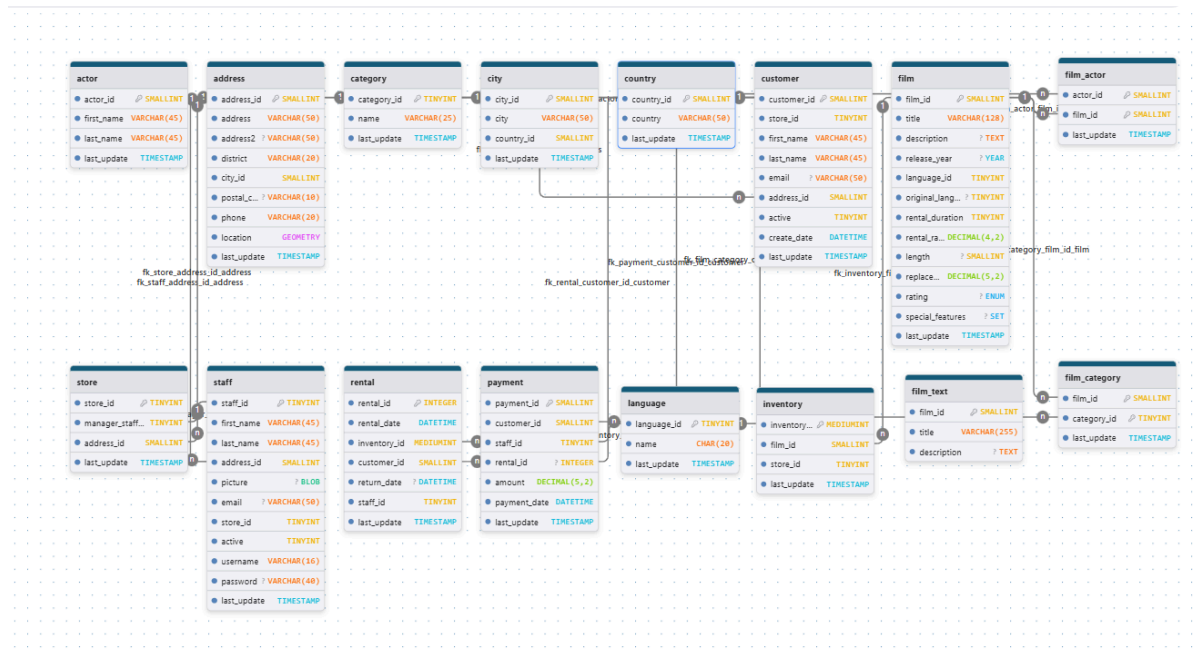
Several operational tables were identified as critical for analytical purposes. The rental table represents the core business transaction because it records each movie rental made by customers. The payment table stores revenue-related information required for financial analysis. Other important tables include customer, film, inventory, store, staff, and category tables, which provide descriptive information used in dimensional analysis.

The inventory table acts as a bridge between films and stores by tracking available movie copies in each branch. Additionally, film_category connects films with their corresponding categories, enabling category-based analysis.

Although the OLTP schema is efficient for operational processing, generating analytical reports requires multiple joins across normalized tables, which can reduce query performance. Therefore, a dimensional data warehouse model is necessary to simplify reporting and improve analytical efficiency.

Table Name	Description
rental	Stores movie rental transactions
payment	Stores customer payment transactions
customer	Contains customer information
film	Contains movie details
inventory	Tracks film copies in stores
store	Contains store branch information
staff	Stores employee information
category	Defines movie categories

2.1 Original OLTP EER Diagram :



3. Business Questions and Analytical Requirements :

The primary goal of the proposed data warehouse is to support analytical reporting and business decision-making for the movie rental company. Based on the analysis of the OLTP schema, several important business questions were identified. These questions focus on rental activity, revenue generation, customer behavior, film popularity, store performance, staff efficiency, and time-based business trends.

The following analytical questions should be supported by the data warehouse:

Business Question	Importance for Decision-Making
Which films are rented most frequently?	Helps identify the most popular films and customer preferences.
Which films generate the highest revenue?	Supports profitability analysis and pricing decisions.
Which film categories are most popular?	Helps management understand customer interests and optimize inventory.
Which stores generate the highest number of rentals?	Enables comparison of store performance and operational efficiency.
Which stores generate the highest revenue?	Helps evaluate branch profitability and business growth opportunities.
Which customers rent the most films?	Identifies loyal customers for marketing and customer retention strategies.
Which customers generate the highest revenue?	Helps recognize high-value customers and improve customer relationship management.

How does rental activity change over time?	Supports trend analysis and seasonal business planning.
How does revenue change by month, quarter, or year?	Helps management monitor financial performance over time.
Which staff members process the highest number of rentals or payments?	Assists in evaluating employee productivity and performance.
Which cities or countries generate the highest customer activity?	Supports location-based business analysis and expansion planning.
What is the average rental duration for different films or categories?	Helps evaluate customer rental behavior and optimize rental policies.
Which films are returned late most often?	Supports late-return monitoring and policy improvement.
How does store performance differ by location?	Enables regional performance comparison and strategic planning.
What are the most active customer locations?	Helps identify important market areas and customer concentration.

These business questions provide the foundation for designing the dimensional data warehouse model. The identified analytical requirements will guide the selection of fact tables, dimension tables, measures, and data granularity in the proposed warehouse design.

4. Dimensional Model Design :

The dimensional model was designed based on the identified business requirements and analytical questions. The purpose of the model is to support business intelligence, reporting, and decision-making for the movie rental company.

The proposed dimensional model focuses on two major business processes within the OLTP system:

- Payment processing
- Customer rental activity

These processes were selected because they represent the core operational and analytical activities of the business.

4.1 Main Business Processes

The following business processes were identified as the most important analytical areas:

Business Process	Description
Payment Processing	Tracks customer payments and business revenue
Customer Rental Activity	Tracks customer rental behavior and activity

4.2 Fact Tables

Fact_Payment was selected because payment transactions represent the primary source of business revenue. This fact table enables financial analysis such as revenue trends, store profitability, customer spending behavior, and payment activity over time. Since payment transactions are measurable, transaction-oriented, and frequently generated, they are highly suitable for dimensional analysis.

Fact_Customer_Activity was selected to support customer behavior analysis and rental activity monitoring. This fact table allows the business to analyze customer rental patterns, rental frequency, late returns, and customer engagement across different films, stores, and time periods. Customer activity is a core operational process that provides valuable insights for customer retention and business performance evaluation.

Fact_Payment

The Fact_Payment table stores information related to customer payment transactions.

Grain

The grain of the Fact_Payment table is defined as:

One record per payment transaction made by a customer for a rental.

Measures

Measure	Description
payment_amount	Amount paid by the customer
payment_count	Number of payment transactions

Foreign Keys

Fact_Payment contains the following foreign keys: customer_key linked to Dim_Customer, film_key linked to Dim_Film, store_key linked to Dim_Store, staff_key linked to Dim_Staff, and date_key linked to Dim_Date. These relationships allow the fact table to support multidimensional analysis across customers, films, stores, staff members, and time periods.

Fact_Customer_Activity

The Fact_Customer_Activity table captures customer rental activity and behavior.

Grain

One record per customer rental activity transaction.

Measures

Measure	Description
rental_count	Total number of rentals
rental_duration	Rental duration in days
late_return_days	Number of days returned late

Foreign Keys

The Fact_Payment table contains several foreign keys that connect the fact table to related dimension tables within the dimensional model. The customer_key links to the Dim_Customer table to support customer-based analysis, while the film_key connects to Dim_Film for film-related reporting. The store_key is linked to Dim_Store to analyze store

performance, and the staff_key connects to Dim_Staff for employee performance evaluation. In addition, the date_key links to Dim_Date to enable time-based analysis such as monthly and yearly revenue trends. These relationships allow the data warehouse to support multidimensional analytical reporting across different business areas.

4.3 Dimension Tables

Several dimension tables were designed to provide descriptive information for analytical reporting.

Dimension Table	Source OLTP Tables	Purpose
Dim_Customer	customer, address, city, country	Customer analysis
Dim_Film	film, film_category, category, language	Film analysis
Dim_Store	store, address, city, country	Store performance analysis
Dim_Staff	staff, store	Employee performance analysis
Dim_Date	Generated in DW	Time-based analysis
Dim_Category	category	Film category analysis
Dim_Location	city, country	Geographic analysis

4.4 Shared Dimensions

Several dimensions are shared between fact tables in order to support integrated analytical reporting across multiple business processes. Shared dimensions allow consistent analysis and reduce data redundancy within the dimensional model.

For example, the Dim_Customer table is shared between Fact_Payment and Fact_Customer_Activity because both business processes involve customer-related analysis. Similarly, Dim_Date is shared to support time-based reporting for both payment and rental activities.

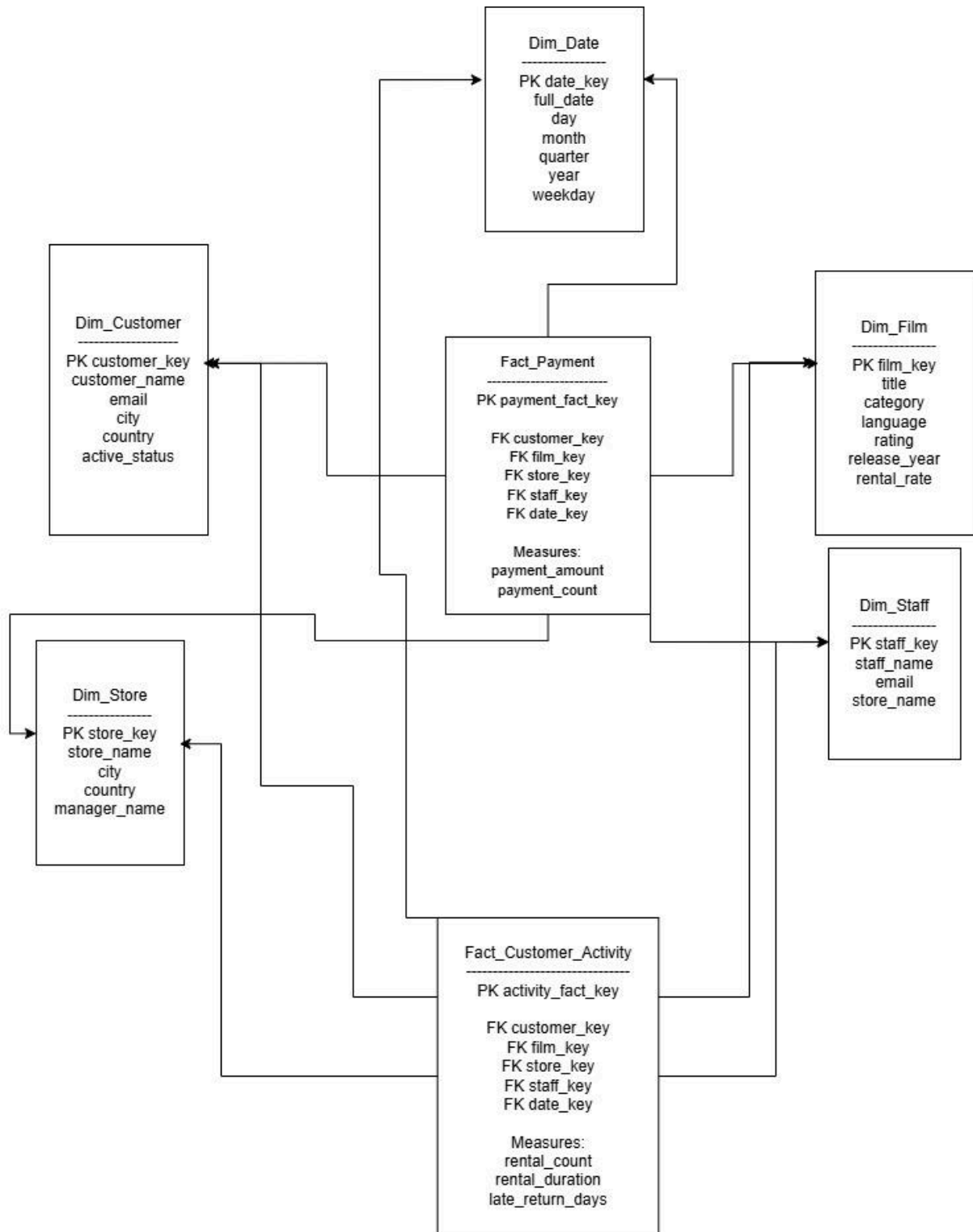
Using shared dimensions improves reporting consistency and enables cross-process business analysis.

Shared Dimension	Used By
Dim_Customer	Fact_Payment, Fact_Customer_Activity
Dim_Film	Fact_Payment, Fact_Customer_Activity
Dim_Store	Fact_Payment, Fact_Customer_Activity
Dim_Staff	Fact_Payment, Fact_Customer_Activity
Dim_Date	Fact_Payment, Fact_Customer_Activity

4.5 Schema Design

The proposed dimensional model follows a **Star Schema design**. The star schema was selected because it simplifies analytical queries, improves query performance, and provides an easy-to-understand structure for business intelligence reporting.

The design contains centralized fact tables connected directly to denormalized dimension tables. This structure is highly suitable for OLAP analysis and supports efficient reporting across multiple business areas.



4.6 Source OLTP Tables

The following OLTP tables are used as data sources for the dimensional model:

Data Warehouse Table	Source OLTP Tables
Fact_Payment	payment, rental
Fact_Customer_Activity	rental, inventory
Dim_Customer	customer, address, city, country
Dim_Film	film, film_category, category, language
Dim_Store	store, address, city, country
Dim_Staff	staff
Dim_Category	category
Dim_Location	city, country

5. ETL Process :

The proposed data warehouse requires an ETL (Extract, Transform, Load) process to move data from the OLTP system into the dimensional warehouse environment. The ETL process ensures that operational data is cleaned, transformed, integrated, and loaded in a format suitable for analytical reporting and OLAP analysis.

5.1 Extract Phase

During the extract phase, data is collected from several operational tables within the OLTP movie rental system. Only tables relevant to analytical reporting are selected for extraction.

Source Table	Purpose
payment	Source for payment transactions and revenue analysis
rental	Source for customer rental activity
customer	Customer information
film	Film details and rental information
inventory	Connects films with stores
store	Store information
staff	Employee information
address	Customer and store addresses
city	Geographic analysis
country	Country-level analysis

category	Film category analysis
film_category	Connects films with categories
language	Film language information

5.2 Transform Phase

During the transformation phase, operational data extracted from the OLTP system is cleaned, integrated, and converted into a format suitable for dimensional analysis.

Several OLTP tables are joined together to create denormalized dimension tables. For example, customer, address, city, and country tables are combined to build the Dim_Customer table, while film, category, film_category, and language tables are integrated to create the Dim_Film dimension.

Data cleansing operations are applied to remove duplicate, missing, or invalid records. Customer and staff first and last names are combined into full names to improve readability and reporting quality.

Operational identifiers from the OLTP system are converted into surrogate keys used within the data warehouse dimensions. Date values such as rental dates and payment dates are transformed into date keys connected to the Dim_Date table.

Several analytical measures are calculated during transformation, including payment amount, rental count, rental duration, and late return days. Late returns are identified by comparing the actual rental duration with the expected rental duration defined in the film table.

Text values such as city names, country names, languages, and status values are standardized to ensure consistency across analytical reports. The transformation process also prepares the warehouse to support future historical tracking of customer, store, staff, and film information.

5.3 Load Phase

After the transformation process is completed, the cleaned and integrated data is loaded into the data warehouse environment.

Dimension tables are loaded before fact tables because both Fact_Payment and Fact_Customer_Activity depend on dimension surrogate keys. Surrogate keys are generated

for all dimension tables including customer, film, store, staff, and date dimensions to improve consistency and analytical performance.

The recommended loading sequence is:

1. Dim_Date
2. Dim_Customer
3. Dim_Film
4. Dim_Store
5. Dim_Staff
6. Fact_Customer_Activity
7. Fact_Payment

The fact tables reference the surrogate keys of the related dimension tables through foreign key relationships. New records are inserted during each ETL cycle, while changed records may be updated according to the selected data maintenance strategy.

Data quality validation checks are applied before and after loading to ensure data accuracy, consistency, and completeness within the warehouse. This loading process ensures that the warehouse supports efficient analytical reporting and business intelligence operations.

6.Data Quality Considerations :

Data quality checks are important to ensure accurate and reliable analytical reporting in the data warehouse. During the ETL process, duplicate and missing records should be identified and handled properly. Important fields such as customer information, payment amounts, rental dates, and film identifiers must contain valid data.

Validation rules should ensure that payment amounts are greater than zero and that return dates are not earlier than rental dates. Text values such as city names, country names, and film categories should also be standardized to maintain consistency across reports.

In addition, referential integrity checks should confirm that all foreign keys in fact tables correctly match existing records in the related dimension tables. These data quality rules help improve the accuracy, consistency, and reliability of the data warehouse.

7.Conclusion :

This project proposed a high-level dimensional data warehouse design for the Sakila movie rental system. The proposed solution transforms the operational OLTP database into a structure optimized for analytical reporting and business intelligence.

The design focused on two major business processes: payment transactions and customer rental activity. Based on these processes, two central fact tables, Fact_Payment and

Fact_Customer_Activity, were designed along with several shared dimension tables including customer, film, store, staff, and date dimensions.

A star schema design was selected because it simplifies analytical queries and improves reporting performance. The proposed dimensional model supports important business analysis areas such as revenue analysis, customer behavior analysis, store performance evaluation, film popularity analysis, and time-based trend analysis.

In addition, a high-level ETL process was proposed to extract operational data from the OLTP system, transform it into a clean and integrated format, and load it into the data warehouse environment. Data quality rules were also introduced to ensure data consistency, accuracy, and reliability.

Overall, the proposed data warehouse design provides a scalable and efficient analytical environment that supports business decision-making, improves reporting capabilities, and enables management to gain valuable insights from operational data.